

Decent Coaching Classes
VIII SB
Mind Your Math - IV

Subject: Mathematics Chapters: 4, 5, 6, 7

Marks: 30

Time: 1 hour

Instructions:

- i. All questions are compulsory unless a choice is given.
- ii. Numbers to the right of questions indicate full marks.
- iii. Draw neat, labelled figures wherever necessary.
- iv. Use of a calculator is not allowed.

Q.1 (A) Choose the correct alternative.

(4)

1. The point of concurrence of the altitudes of a triangle is called the _____.
(A) centroid (B) orthocentre (C) incentre (D) circumcentre
2. $(a - b)^3 =$ _____.
(A) $a^3 + b^3$
(B) $a^3 - b^3$
(C) $a^3 - 3a^2b + 3ab^2 - b^3$
(D) $a^3 + 3a^2b + 3ab^2 + b^3$
3. $a^3 + b^3 =$ _____.
(A) $(a + b)(a^2 - ab + b^2)$
(B) $(a - b)(a^2 + ab + b^2)$
(C) $(a + b)(a^2 + ab + b^2)$
(D) $(a - b)(a^2 - ab + b^2)$
4. If x varies inversely as y, then _____ is constant.
(A) $x + y$ (B) $x - y$ (C) xy (D) x/y

Q.1 (B) Solve any ONE of the following.

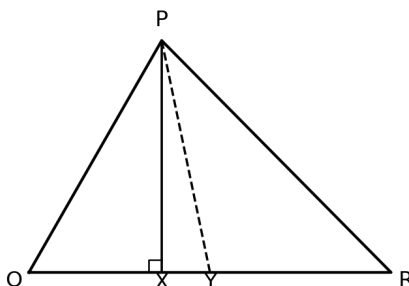
(1)

1. Fill in the blank: The orthocentre of a right-angled triangle lies at the _____.
2. Fill in the blank: If x varies inversely as y, the equation of variation is written as $x =$ _____.

Q.2 Solve any FOUR of the following sub-questions.

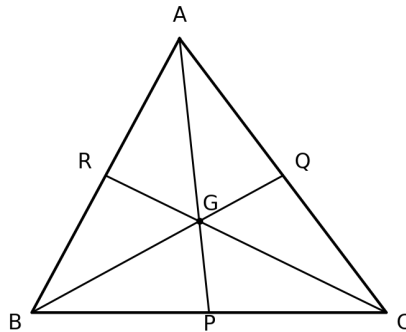
(8)

1. Expand: $(2x - 3y)^3$
2. Factorise: $3x^2 + 14x + 15$
3. Simplify: $(x^2 - 9y^2) \div (x - 3y)$
4. If m varies directly as n, and $m = 154$ when $n = 7$, find the value of m when $n = 14$.
5. In the figure given below, seg PX and seg PY are drawn on side QR of $\triangle PQR$, where X is the foot of the perpendicular from P and Y is the midpoint of QR. Name the altitude and the median of $\triangle PQR$ from vertex P.



Q.3 Solve any THREE of the following sub-questions.**(9)**

1. Draw an obtuse-angled $\triangle STV$. Draw its medians and show the centroid by G.
2. In the figure given below, G is the centroid of $\triangle ABC$. Seg AP, seg BQ and seg CR are its medians. If $l(AG) = 8.4$ cm and $l(BQ) = 9$ cm, find $l(GP)$ and $l(BG)$.



3. Expand: $(2p + 3q + 4)^2$
4. Factorise: $3p^2 - 3p - 60$
5. Simplify: $(3x + 2y)^3 - (3x - 2y)^3$

Q.4 Solve any TWO of the following sub-questions.**(8)**

1. (a) Draw a right-angled $\triangle XYZ$. Draw all of its altitudes and name the orthocentre O. State the location of O with reason.
(b) Draw an equilateral triangle. Find its centroid G. State the location of G.
2. Eighteen workers can build a wall in 40 hours.
(a) Find the constant of variation and write the equation of variation.
(b) How many workers will be required to build the same wall in 24 hours?
3. (a) Factorise: $27p^3 + 8q^3$
(b) Simplify: $(a^2 - 49) \div (a^2 + 3a - 28)$

— *End of Paper* —